Tutorial 3: Partial Wave Expansions.

[Schultz]

Bessel function of first & second kinds:

j_ℓ , n_ℓ [n_ℓ : "Neumann Functions"]

Generally, for radial PDEs of wavefunctions,

$$\psi_{k\ell m}(\vec{r}) \longrightarrow R_{k\ell}(r)\, Y_\ell^m(\theta,\varphi)$$

where $R_{k\ell}(r)$ obeys the Bessel differential equation

$$\boxed{r^2 \frac{d^2 R_{k\ell}}{dr^2} + 2r\frac{dR_{k\ell}}{dr} + \left(k^2 r^2 - \ell(\ell+1)\right) R_{k\ell} = k^2 R_{k\ell}}$$

and whose solution is a linear combination of Bessel/Neumann functions

$$\boxed{R_{k\ell}(r) = \alpha\, j_\ell(kr) + \beta\, n_\ell(kr)}$$

where

$$\boxed{\begin{aligned} j_\ell(kr) &\sim \frac{1}{kr}\sin\left(kr - \frac{\ell\pi}{2}\right) \\ n_\ell(kr) &\sim \frac{1}{kr}\cos\left(kr - \frac{\ell\pi}{2}\right) \end{aligned}}$$

It is useful to define the Hankle function

$$\boxed{h_\ell^{(1)}(kr) = j_\ell(kr) + i n_\ell(kr)}$$

which has the asymptotic form $\boxed{h_\ell^{(1)}(kr) \sim (-i)^{\ell+1} \frac{e^{ikr}}{r} \quad [kr \gg 0]}$

and thus the wavefunction can be written as

$$\psi_{k\ell m}(\vec{r}) = \sum_{\ell,m} c_{\ell,m} \, h_\ell^{(1)}(r) \, Y_\ell^m(\theta,\varphi)$$

change of basis into h_ℓ / Y_ℓ^m space.

If scattering independent of φ-angle, can just set $m=0$:

$$\psi_{k\ell 0}(\vec{r}) = \sum_\ell c_\ell \, h_\ell^{(1)}(r) \sqrt{\frac{2\ell+1}{4\pi}} \, P_\ell(\cos\theta).$$

ψ must be of the form, with $f_k(\theta,\varphi)$ accounted for, as

$$\boxed{\psi = A\left\{ e^{ikz} + k\sum_{\ell=0}^{\infty} i^{\ell+1}(2\ell+1)\, a_\ell \, h_\ell^{(1)}(kr)\, P_\ell(\cos\theta)\right\}}$$

$$f_k(\theta) \sim \sum_{\ell=0}^{\infty} a_\ell (2\ell+1) P_\ell(\cos\theta)$$

and with Rayleigh's formula $\quad e^{ikz} = \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) \, j_\ell(kr) \, P_\ell(\cos\theta)$

we can rewrite this much more simply:

$$\boxed{\psi = A\sum_{\ell=0}^{\infty} i^\ell (2\ell+1) P_\ell(\cos\theta)\left[j_\ell(kr) + k i a_\ell h_\ell^{(1)}(kr)\right]}$$

a_ℓ: partial wave amplitude.

Problems:

Q1) Recall $\psi_{k\ell}(\vec{r}) = A \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) \left[j_\ell(kr) + ika_\ell h_\ell^{(1)}(kr) \right] P_\ell(\cos\theta)$.

2) Show that, if we have the wave function

$$\psi(\vec{r}) = A \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) \left[b_\ell h_\ell^{(1)}(kr) + c_\ell h_\ell^{(2)}(kr) \right) P_\ell(\cos\theta),$$

then $|b_\ell|^2 = |c_\ell|^2$ [ie, probability must be conserved].

Continuity: $\frac{\partial}{\partial t} |\psi|^2 + \nabla \cdot \vec{j} = 0$.

The time-dependent wave function is thus

$$\psi(\vec{r},t) = A \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) P_\ell(\cos\theta) \left[b_\ell h_\ell^{(1)}(kr) + c_\ell h_\ell^{(2)}(kr) \right] e^{-iE_k t/\hbar}$$

because all states with the same k are degenerate.

Therefore $\frac{\partial |\psi|^2}{\partial t} = 0$ $\quad [e^{-it} \cdot e^{it} = 1]$ $\quad$ and $\int \nabla \cdot \vec{j} = \oint \vec{j} \cdot d\vec{A}$.

For an area element $r^2 \sin\theta \, d\theta \, d\varphi$, we have

$$= \frac{i\hbar}{2m} \int_0^\pi \int_0^{2\pi} (\psi \nabla \psi^* - \psi^* \nabla \psi) \, r^2 \hat{r} \sin\theta \, d\theta \, d\varphi$$

$$= \frac{i\pi\hbar}{m} \int_0^\pi (\psi \nabla \psi^* + \text{c.c.}) \, r^2 \hat{r} \sin\theta \, d\theta.$$

Plugging in ψ, we have

$$= \frac{\pi i \hbar}{m} \int_0^\pi \Big[\sum_{\ell=0}^{\infty} \sum_{\ell'=0}^{\infty} i^\ell (2\ell+1) \left[b_\ell h_\ell^{(1)}(kr) + c_\ell h_\ell^{(2)}(kr) \right] P_\ell(\cos\theta)$$

$$\cdot (-i)^{\ell'} (2\ell'+1) \left[b_{\ell'}^* \partial_r h_{\ell'}^{(2)}(kr) + c_\ell^* \partial_r h_{\ell'}^{(1)}(kr) \right] P_{\ell'}(\cos\theta) + c.c. \Big] \hat{r}\, r^2 \sin\theta \, d\theta$$

Using $\int_0^\pi P_\ell(\cos\theta) P_{\ell'}(\cos\theta) \sin\theta\, d\theta = \frac{2\delta_{\ell\ell'}}{2\ell+1}$,

$$\oint \vec{J} \cdot d\vec{A} = \frac{\pi i \hbar}{m} \Big[i^\ell (2\ell+1) \left[b_\ell h_\ell^{(1)}(kr) + c_\ell h_\ell^{(2)}(kr) \right] \cdot (-i)^\ell \cancel{(2\ell+1)} \left[b_\ell^* \partial_r h_\ell^{(2)}(kr) + c_\ell^* \partial_r h_\ell^{(1)}(kr) \right] - c.c. \Big] r^2$$

implicit sum

$$= \frac{i\pi\hbar r^2}{m} \Big[(2\ell+1) \Big[|b_\ell|^2 h_\ell^{(1)}(kr)\, \partial_r h_\ell^{(2)}(kr) + c_\ell b_\ell^* h_\ell^{(2)}(kr)\, \partial_r h_\ell^{(2)}(kr)$$

$$+ b_\ell c_\ell^* h_\ell^{(1)}(kr)\, \partial_r h_\ell^{(1)}(kr) + |c_\ell|^2 h_\ell^{(2)}(kr)\, \partial_r h_\ell^{(1)}(kr) - c.c. \Big]$$

Note that, upon subtracting off the complex conjugate, the cross terms

$$c_\ell b_\ell^* h_\ell^{(2)}(kr)\, \partial_r h_\ell^{(2)}(kr) + b_\ell c_\ell^* h_\ell^{(1)}(kr)\, \partial_r h_\ell^{(1)}(kr)$$

are real, so they go to 0. We obtain

$$\oint \vec{J} \cdot d\vec{A} = \frac{2\pi i \hbar}{m} \Big[(2\ell+1) \underbrace{\left[|b_\ell|^2 - |c_\ell|^2 \right]}_{\boxed{=0}} \underbrace{\left[h_\ell^{(1)}(kr)\, \partial_r h_\ell^{(2)}(kr) - h_\ell^{(2)}(kr)\, \partial_r h_\ell^{(1)}(kr) \right]}_{\neq 0} \Big]$$

since $\frac{\partial \rho}{\partial t} = 0$.

Thus $|b_\ell|^2 = |c_\ell|^2$.

To be more specific, however,

$$h_\ell^{(1)}(kr)\,\partial_r h_\ell^{(2)}(kr) - h_\ell^{(2)}(kr)\,\partial_r h_\ell^{(1)}(kr) = -\frac{2i}{kr^2}$$

since this is just an application of the Sturm-Liouville theorem

$$(py')' - qy + \lambda r y = 0$$

the Wronskian is $W(x) = \frac{c}{p(x)}$.

Thus $c = -\frac{2i}{k}$, $p(x) \to p(r) = r$, and we find

$$\oint \vec{j}\cdot d\vec{A} = \frac{4\pi\hbar}{m}\sum_{\ell=0}^{\infty}(2\ell+1)\underbrace{\left[|b_\ell|^2 - |c_\ell|^2\right]}_{=0}.$$

b) What are the implications of (a) on the wave function

$$\psi = A\sum_{\ell=0}^{\infty} i^\ell (2\ell+1)\left[j_\ell(kr) + i a_\ell k\, h_\ell^{(1)}(kr)\right] P_\ell(\cos\theta) \ ?$$

Determine the partial wave amplitude a_ℓ as a phase shift.

First, we have that $j_\ell(kr) = \frac{1}{2}\left[h_\ell^{(1)}(kr) + h_\ell^{(2)}(kr)\right]$ hence

$$j_\ell(kr) + i a_\ell k h_\ell^{(1)}(kr) \to \frac{1}{2}h_\ell^{(2)}(kr) + \left(i a_\ell k + \frac{1}{2}\right)h_\ell^{(1)}(kr).$$

and since $\left|\frac{1}{2}\right|^2 = \left|i a_\ell k + \frac{1}{2}\right|^2$ are of equal magnitude,

we can set $\frac{1}{2} + i a_\ell k = \frac{1}{2}e^{2i\delta_\ell}$, thus arriving at $\boxed{a_\ell = \frac{1}{2ik}\left(e^{2i\delta_\ell} - 1\right)}$

$$= \frac{1}{k} e^{i\delta_\ell} \sin(\delta_\ell).$$

c) determine cross section

This yields the total cross-section

$$\sigma = 4\pi \sum_{\ell=0}^{\infty} (2\ell+1)|a_\ell|^2$$

$$= 4\pi \sum_{\ell=0}^{\infty} (2\ell+1) \sin^2(\delta_\ell) \cdot \frac{1}{k^2}$$

$$= \frac{4\pi}{k^2} \underbrace{\sum_{\ell=0}^{\infty} (2\ell+1) \sin^2(\delta_\ell)}_{\frac{1}{k} \mathrm{Im}(f_k(0))}.$$

Q2) Consider the infinite potential barrier $V(r) = \begin{cases} \infty & r < a \\ 0 & r \geq a \end{cases}$.

a) Compute the partial wave shift δ_ℓ.

From last week,

$$\psi = A \sum_{\ell=0}^{\infty} i^\ell (2\ell+1) P_\ell(\cos\theta) \left[j_\ell(ka) + k i a_\ell h_\ell^{(1)}(ka) \right] = 0$$ at $r=a$ boundary

hence $a_\ell = \frac{-j_\ell(kr)}{k i h_\ell^{(1)}(kr)} = \frac{i j_\ell(ka)}{k h_\ell^{(1)}(ka)}$

$$= \frac{1}{2ik}\left(e^{2i\delta_\ell} - 1\right).$$

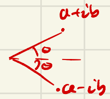
Thus $$-\frac{j_\ell(ka)}{h_\ell^{(1)}(ka)} = \frac{1}{2}\left(e^{2i\delta_\ell}-1\right)$$

$$\Rightarrow \quad e^{2i\delta_\ell} = 1-\frac{2j_\ell(ka)}{h_\ell^{(1)}(ka)}$$

$$= \frac{h_\ell^{(1)}(ka)-2j_\ell(ka)}{h_\ell^{(1)}(ka)}$$

$$= \frac{j_\ell(ka)+in_\ell(ka)-2j_\ell(ka)}{h_\ell^{(1)}(ka)}$$

$$= -\frac{h_\ell^{(2)}(ka)}{h_\ell^{(1)}(ka)} \quad \sim -\frac{a-ib}{a+ib}$$



$\cdot(-1)$

$$\sim -\frac{Ae^{i\theta}}{Ae^{-i\theta}} = -e^{2i\theta}$$

$$= e^{i\pi-2i\theta}$$

$$\Rightarrow 2\delta_\ell = \pi - 2\,\mathrm{Arg}\left(h_\ell^{(1)}(ka)\right)$$

$$\Rightarrow \boxed{\delta_\ell = \frac{\pi}{2} - \mathrm{Arg}\left(h_\ell^{(1)}(ka)\right)}$$

$$\mathrm{Arg}\left(h_\ell^{(1)}(ka)\right) \sim \tan\left(\frac{n_\ell(ka)}{j_\ell(ka)}\right).$$

b) Determine δ_ℓ for low energies $ka \ll 1$.

Note first that for $|z| \ll 1$,

$$j_\ell(z) \sim \frac{z^\ell 2^\ell \ell!}{(2\ell)!}$$

$$n_\ell(z) \sim -\frac{(2\ell+1)!}{z^{\ell+1}2^\ell \ell!}$$

Thus $$\delta_\ell = \frac{\pi}{2} - \arctan\left(\frac{n_\ell(ka)}{j_\ell(ka)}\right)$$

$$= \frac{\pi}{2} - \arctan\left(-\frac{(2\ell+1)!}{(ka)^{\ell+1}2^\ell \ell!}\cdot\frac{(2\ell)!}{(ka)^\ell 2^\ell \ell!}\right)$$

$$= \frac{\pi}{2} + \arctan\left(\frac{(2\ell+1)!\,(2\ell)!}{(ka)^{2\ell+1}\,2^{2\ell}(\ell!)^2}\right)$$

$$= \frac{\pi}{2} - \arctan\left(\frac{(ka)^{2l+1}}{(2l-1)!}\frac{2^{2l}(l!)^2}{(2l)!}\right)$$

$$\simeq \frac{\pi}{2} - \underbrace{\frac{(ka)^{2l+1}\, 2^{2l}(l!)^2}{(2l-1)!\,(2l)!}}_{\text{Leading order term.}}$$

Q3) Consider the potential well

$$V(r) = \begin{cases} -V_0 & r < a \\ 0 & r \geq a \end{cases} .$$

a) Compute the partial wave shifts δ_l.

The schrödinger equation is given by

$$-\frac{\hbar^2}{2mr^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}R_{kl}\right) + V_{eff}(r)\,R_{kl} = \frac{\hbar^2k^2}{2m}R_{kl}$$

and $V_{eff}(r) = V(r) + \dfrac{\hbar^2 l(l+1)}{2mr^2}$.

In the $r > a$ region, the wave is free, hence

$$\psi_{out} = A\sum_{l=0}^{\infty} i^l\frac{(2l+1)}{2}\Big[\underbrace{h_l^{(1)}(kr)e^{2i\delta_l}}_{\substack{\text{outgoing wave}\\ \text{w/ phase shift}}} + \underbrace{h_l^{(2)}(kr)}_{\text{incoming}}\Big]P_l(\cos\theta)$$

and

$$\psi_{in} = B\sum_{l=0}^{\infty} i^l(2l+1)\, C_l\, j_l(k_0 r)\, P_l(\cos\theta)$$

Some change of basis in $j_l(kr)$ which satisfies $V = -V_0$ SE.

k_0: restricted momentum.

The wavefunctions must be continuous at $r=a$:

$$A\, i^{\ell} \frac{(2\ell+1)}{2} \left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right] P_\ell(\cos\theta)$$

$$= B\, i^{\ell'} (2\ell'+1)\, c_{\ell'}\, j_{\ell'}(k_0 a)\, P_{\ell'}(\cos\theta)$$

$$\Rightarrow \int_0^\pi B\, i^{\ell'} \cancel{(2\ell'+1)}\, j_{\ell'}(k_0 a)\, c_{\ell'}\, \underbrace{\cancel{P_{\ell'}(\cos\theta)} P_{\ell'}(\cos\theta) \sin\theta d\theta}_{\frac{2}{2\ell'+1}}$$

$$= A\, i^{\ell} \frac{(2\ell+1)}{2} \left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right] \int_0^\pi \sin\theta d\theta\, P_\ell(\cos\theta) P_{\ell'}(\cos\theta)$$

$$= A\, i^{\ell} \frac{\cancel{(2\ell+1)}}{\cancel{2}} \left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right] \frac{2\delta_{\ell\ell'}}{\cancel{2\ell+1}}$$

$$2 \cdot B\, \cancel{i^{\ell}}\, j_\ell(k_0 a)\, c_\ell = A \cancel{i^{\ell}} \left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right]$$

$$2B \cdot j_\ell(k_0 a) \cdot c_\ell = A \left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right)$$

and the derivatives continuous:

$$2B\, k_0\, j_\ell'(k_0 a)\, c_\ell = A k \left[h_\ell^{(1)'}(ka) e^{2i\delta_\ell} + h_\ell^{(2)'}(ka)\right]$$

then, dividing,

$$k \frac{\left[h_\ell^{(1)'}(ka) e^{2i\delta_\ell} + h_\ell^{(2)'}(ka)\right]}{\left[h_\ell^{(1)}(ka) e^{2i\delta_\ell} + h_\ell^{(2)}(ka)\right]} = k_0 \frac{j'(k_0 a)}{j(k_0 a)}$$

Which can be brutally arranged to find that

$$e^{2i\delta_\ell} = \frac{k_0 j_\ell'(k_0 a) h_\ell^{(2)}(ka) - k j_\ell(k_0 a) h_\ell^{(2)'}(ka)}{k j_\ell(k_0 a) h_\ell^{(1)'}(ka) - k_0 j_\ell'(k_0 a) h_\ell^{(1)}(ka)}.$$

Consult tut notes. This gets insane.